

TMS Architecture Sampler

Block 4 Final Report

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1. Problem and Motivation

Why this project exists

Training a GNN to reason about BEV thermal management system design requires a large, diverse corpus of valid TMS graphs to label and learn from. A valid TMS graph here means one that satisfies a set of operating constraints: each component type has a required degree, the graph must be weakly connected, and physical flow requirements would impose further constraints.

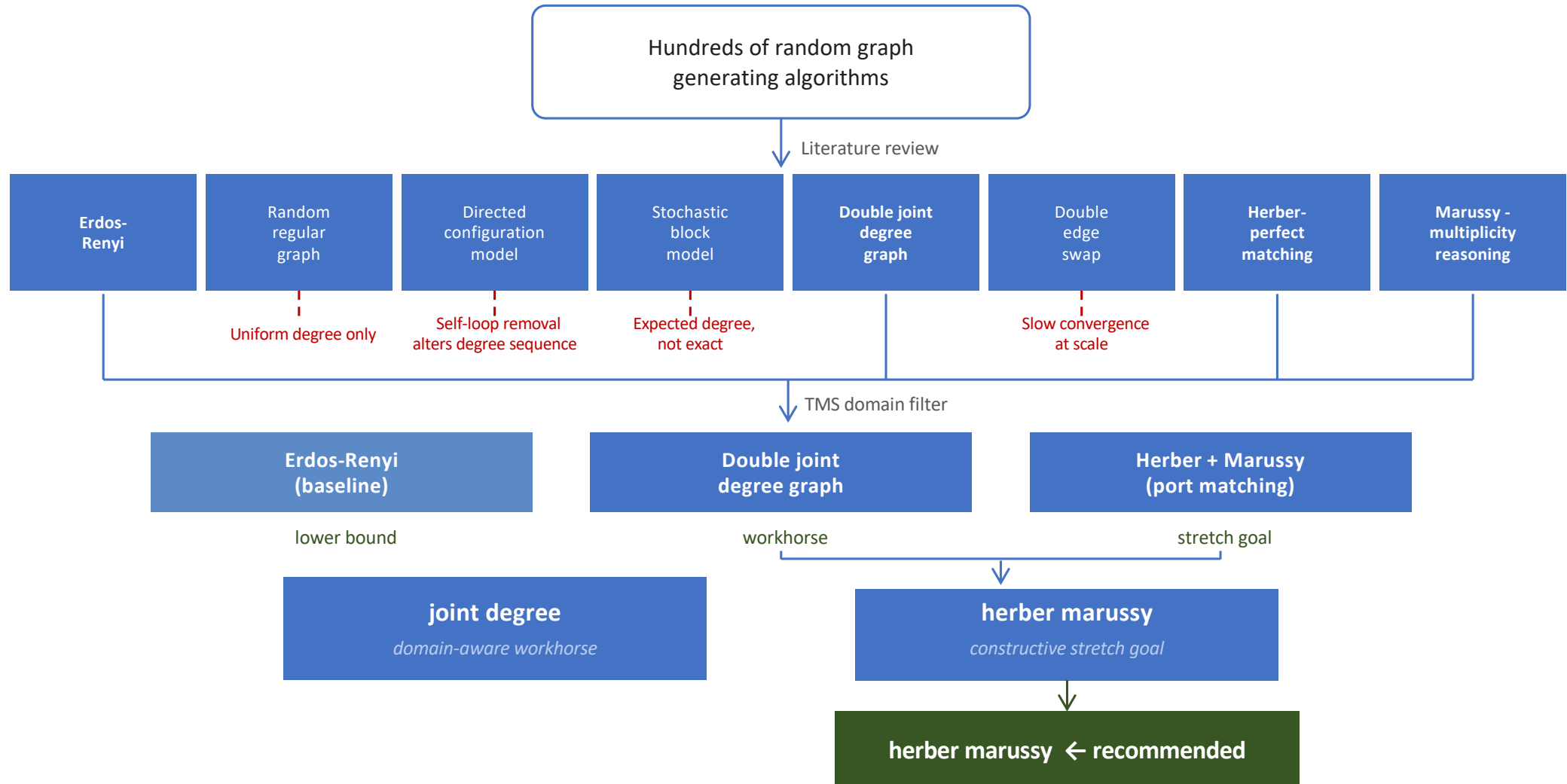
Why not just use a naive random graph generator?

Random graph algorithms like Erdos-Renyi have no awareness of these constraints. They place edges randomly and assign component labels post-hoc, so almost none of what they generate is actually valid, in practice around 2 in every 1,000 graphs pass the TMS feasibility check. This makes naive random sampling impractical as a source of training data, a sampler that enforces structural constraints during generation is needed. Brute-force enumeration of valid graphs is also off the table; the candidate space grows too fast past small component counts to be practical.

What this project does

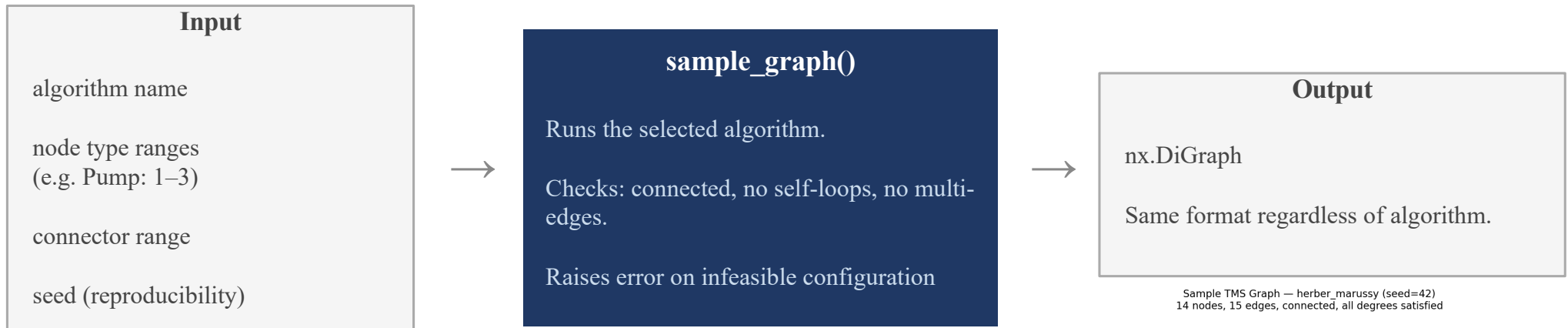
Implements and compares sampling algorithms that generate valid TMS architecture graphs by construction, so the output can actually be used as GNN training data.

2. Algorithm Family and Selection

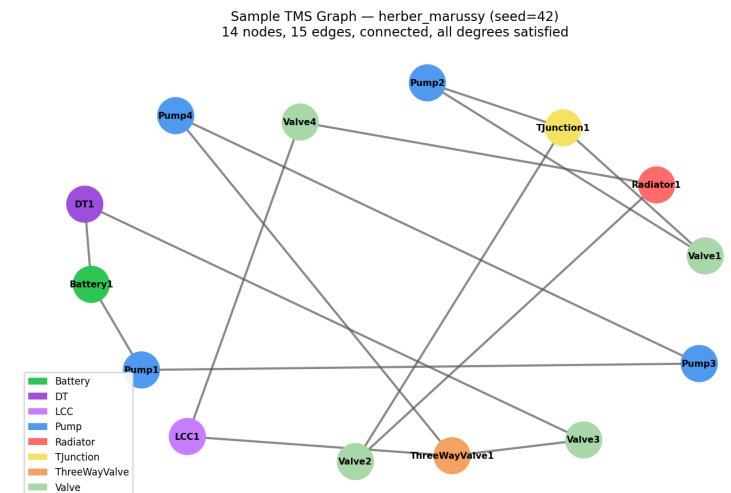


3. Unified Sampler Interface

All four algorithms are called the same way



Having a unified interface means you can swap algorithms without changing anything else in the pipeline. The algorithm is just a parameter.



4. How the Algorithms Work

joint_degree

- Uses NetworkX's `directed_joint_degree_graph`
- The configuration is first translated into a joint degree dictionary by grouping component types by their required degree
- The graph is then realized via stub matching: each node is given a number of "stubs" equal to its degree, and stubs are paired randomly until all are matched.
- Degree constraints are satisfied by construction
- Limitation: the matching step is type-blind since the stub-matching step operates only on degree counts.
- This induces a topological bias and is why the algorithm produces only 679 graph structures out of 5000 samples.

herber_marussy

- Constructive approach from Herber, Guo, and Allison (2017), extended with incremental backtracking from Marussy et al. (IEEE TSE, 2022).
- A port pool is constructed by enumerating one port per degree slot per node.
- A perfect matching is then sought over this pool: ports are paired randomly subject to no-self-loop and no-multi-edge constraints.
- If the algorithm gets stuck and no valid match exists for an unmatched port, it backtracks to the last valid checkpoint and retries, up to 20 attempts.
- Because ports are typed from the start, the matching is inherently type-aware. This is what produces 4,067 distinct graph structures out of 5,000 samples, compared to 679 for `joint_degree`.
- Limitation: 6.6% of generated graphs are disconnected and need to be filtered downstream.

5. Comparative Evaluation -Validity and Speed

Metric	joint_degree	herber_marussy
Generation success	100% (5,000/5,000)	100% (5,000/5,000)
TMS feasibility	100%	100%
Connectivity	100%	93.4%
Mean nodes $\pm \sigma$	27.5 ± 4.7	27.4 ± 4.7
Mean edges $\pm \sigma$	33.8 ± 6.3	33.6 ± 6.3
Mean density $\pm \sigma$	0.0959 ± 0.0190	0.0961 ± 0.0190
Time to generate (5,000 graphs)	2.82 s (0.56 ms/graph)	4.41 s (0.88 ms/graph)
Relative cost	1.0 $\times$ (baseline)	$\sim 1.57\times$

Both algorithms achieve 100% feasibility. The key difference is connectivity: herber_marussy produces disconnected graphs 6.6% of the time, which would need to be filtered out downstream. It is also 1.57 $\times$ slower, though both are fast in absolute terms.

5. Comparative Evaluation -Structural Diversity

Metric	joint_degree	herber_marussy
Edge count σ	6.283	6.296
Distinct JDMs (of 5,000)	679	4,067
Assortativity mean $\pm \sigma$	-0.5520 ± 0.2072	-0.0612 ± 0.1592
Same-type edge fraction	4.6%	11.6%
Distinct topologies(in 5000 samples)	679	4067

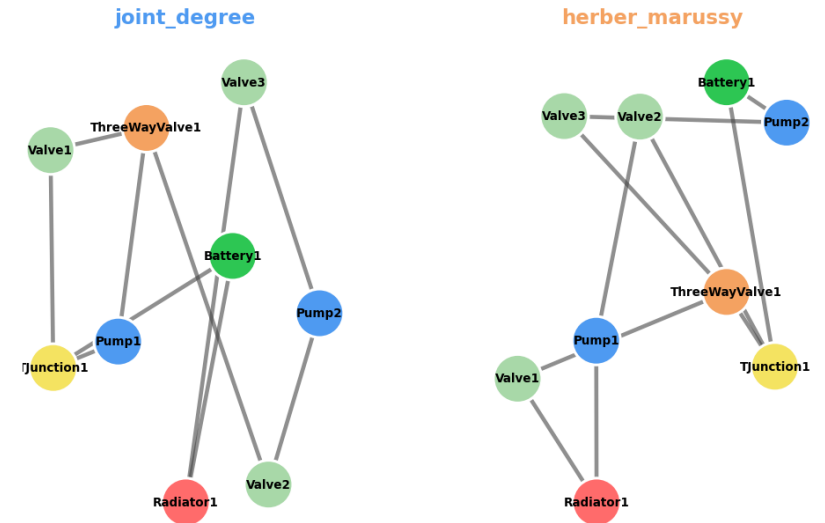
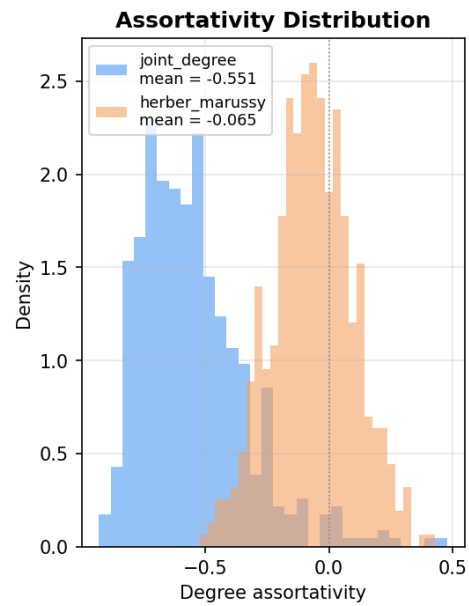
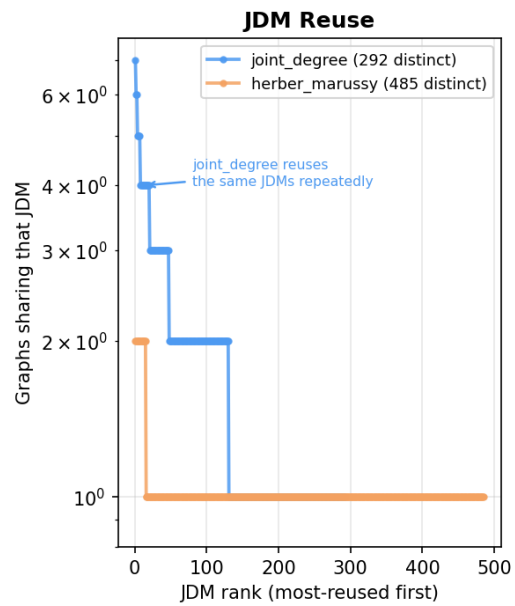
What this means

The edge count standard deviation is almost identical, so aggregate density is not the point. The real difference is in joint degree matrix (JDM) diversity: joint_degree produces only 679 structurally distinct graphs out of 5,000, while herber_marussy produces 4,067. That is a $6\times$ difference in topological variety.

The assortativity numbers iterates the same point differently. joint_degree tends toward strongly disassortative graphs (high-degree nodes connecting to low-degree nodes), which reflects the linear-chain topology bias from stub matching. herber_marussy produces a much wider range of mixing patterns.

Diversity lives in the edges, not the node counts

Same node composition — different edge wiring



1x Battery | 2x Pump | 1x Radiator | 1x TJunction | 1x ThreeWayValve | 3x Valve

6. Stratification Layer

Without stratification, any sampler will skew toward the most common graph structures. The stratification layer wraps any algorithm and resamples until each bin in a target distribution hits its quota. Bins are defined by circuit rank -the number of independent cycles in the graph, computed as $E - V + C$. Higher rank means more parallel flow paths.

Results -3 bins, 20 graphs/bin (60 total target)

Circuit rank bin	joint_degree achieved	herber_marussy achieved	Notes
(2, 6) low rank	18 / 20	17 / 20	Hit 800-attempt budget, still missed quota for both
(6, 8) modal range	20 / 20	20 / 20	Modal range, both hit quota cleanly
(8, 12) high rank	20 / 20	17 / 20	HM falls short
Total	58 / 60	54 / 60	

The low-rank bin is the hard case. Graphs with circuit rank 2–6 are structurally rare under this configuration because the modal circuit rank is 7–8. Neither algorithm can fill that bin within the attempt budget, which is itself a useful finding -it tells you something about the configuration's natural graph distribution.

7. Partial-Feasibility Screen

The screen applies three rules in sequence and stops at the first failure. R2 (rank cap) is the active filter for both algorithms. R1 catches herber_marussy's disconnected graphs. R3 catches nothing because both algorithms already produce feasible graphs.

Rule	Description	joint_degree rejected	herber_marussy rejected
R1	Graph must be weakly connected	0 (0.0%)	70 (7.0%)
R2	Circuit rank ≤ 8	311 (31.1%)	293 (29.3%)
R3	Full feasibility check	0 (0.0%)	0 (0.0%)
—	Passed screen	689 (68.9%)	637 (63.7%)

Note: time savings relative to the full downstream evaluator were not measured. The screen module does not benchmark against the directed evaluation pipeline.

Circuit rank ≤ 8 is a domain constraint that keeps the corpus within a realistic range of thermal circuit complexity. Architectures with more than 8 independent cycles are considered too topologically complex for the current scope.

8. Package Status

```
tms_sampler/  
  __init__.py      sample_graph()  
  algorithms/  
    erdos_renyi.py  
    joint_degree.py  
    herber.py  
    herber_marussy.py  
  stratify.py      shared wrapper  
  screen.py        pre-filter  
  evaluate.py      full evaluator  
  tests/  
    test_sampler.py 24 tests, all pass  
  notebooks/  
    final_demo.ipynb exists, not run  
  pyproject.toml  
  README.md
```

Test coverage

Algorithm / area	Tests
erdos_renyi	5
joint_degree	6 (incl. connectivity test)
herber	5
herber_marussy	5
Stratification, screening, beta sweep	3
Total	24 -all pass

9. Recommendation

Use herber_marussy as the primary generator for the downstream GNN training pipeline.

The primary reason is structural diversity. herber_marussy produces 4,067 distinct joint degree matrices out of 5,000 graphs; joint_degree produces 679. For training a GNN that needs to generalize across different TMS topologies, that 6× difference in variety matters more than the small speed and connectivity differences.

The trade-offs are real and worth stating. herber_marussy is 1.57× slower than joint_degree (0.88 ms vs 0.56 ms per graph), and 6.6% of its graphs fail the connectivity check and need to be filtered. At 5,000 samples that means roughly 330 graphs are dropped. The stratification layer can compensate by resampling to hit bin targets.

The other algorithms are not going away.

joint_degree stays in as a fast option for prototyping and for cases where connectivity guarantees matter more than diversity.
erdos_renyi stays in as the domain-agnostic lower bound.

9. What Is Out of Scope and What Comes Next

Out of scope for this block

Direction assignment is not handled here. The sampler produces undirected graphs. Assigning in and out directions to edges is a separate problem handled downstream in Reihaneh's evaluation pipeline.

Node labels are also not assigned at this stage. Labeling is a downstream step.

Screen timing was not benchmarked against the full downstream evaluator. The module would need to be instrumented alongside the directed evaluation pipeline to measure actual time savings.

ruff formatting and final_demo.ipynb execution are pending.

What we would want to do next

Wider stratification experiments would be useful -the current pilot only tested 60 graphs across 3 bins, which is enough to show the method works but not enough to characterize the full configuration space.

Summary

- GNN training for BEV TMS design needs a large, diverse corpus of valid architecture graphs to label and learn from. Valid means satisfying degree constraints, connectivity, and flow topology rules.
- Erdos-Renyi hits only ~0.2% TMS feasibility because it has no degree filters. Brute-force enumeration does not scale. A dedicated sampler is needed.
- Four algorithms were implemented spanning domain-agnostic, domain-aware, and constructive approaches. Two (joint_degree and herber_marussy) are in the final comparison. None assign edge directions.
- A unified interface wraps all four. The algorithm is just a parameter -you can swap it without changing anything else in the pipeline.
- Both final comparison algorithms achieve 100% feasibility at 5,000 samples. herber_marussy produces 4,067 distinct graph topologies vs. 679 for joint_degree, but 6.6% fail connectivity and need filtering. It is 1.57× slower.
- The stratification layer controls corpus distribution by circuit rank. Both algorithms miss the low-rank bin (rank 2–6) even at 800 attempts - that bin is structurally rare under this configuration.
- The partial-feasibility screen applies three rules in order. R2 (circuit rank ≤ 8) rejects around 30% of graphs for both algorithms. R1 catches herber_marussy's disconnected graphs. R3 catches nothing.
- Recommendation: use herber_marussy for topological diversity. joint_degree stays as a fast, connectivity-safe alternative. 24 tests pass.
- Out of scope: direction assignment, node labeling, screen timing benchmarks. Next candidate algorithm: 2K+A / JDAM.